MILITARY SPECIFICATION

RADAR SET AN/APS-116

* This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be an airborne search radar. The primary objective is to detect submarine periscopes within five seconds after exposure and to provide this detection capability under severe sea conditions to ranges of at least 20 n. miles. Additional requirements include performing the functions of submarine snorkel detection, surface plotting, radar navigation and station keeping.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Type Designation	Applicable Paragraph
Power Supply PP-6633/APS-116	3.5.1
Synchronizer-Exciter SN460/APS-116	3.5.2
Transmitter, Radar Set T-1203/APS-116	3.5.3
Receiver-Pulse Compressor R-1747/APS-116	3.5.4

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-B-5087 - Bonding, Electrical, and Lightning Protection, for Aerospace Systems

* COMMENTS OR RECOMMENDATIONS WHICH MAY BE OF USE IN IMPROVING THIS DOCUMENT MAY BE SENT TO THE PREPARING ACTIVITY WHEN A SELF-ADDRESSED STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL (DD FORM 1426) APPEARS AT THE END OF THIS DOCUMENT, OR BY LETTER ADDRESSED TO THE COGNIZANT ACTIVITY.

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MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing Environmental, Aircraft Electronic Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and associated repair parts, preparation for delivery of
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-E-18927	Environmental Systems, Pressurized Aircraft, General Requirements for
MIL-T-23103	Thermal Performance Evaluation, Airborne Electronic Equipment; General Requirement for

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Precedures for Selection and Approval of
AR-8	Versatile Avionics Shop Test System, Avionics Systems Compatibility, General Requirements for
AR-9	Versatile Avionics Shop Test System Test Program, General Requirements for
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for
WR-104	Naval Weapons Requirements-Radar Engineering Design Objective, requirements for

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-463	Definitions and Systems of Units, Electromagnetic Interference Technology
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

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MIL-STD-781 Reliability Tests, Exponential Distribution

MIL-STD-785 Reliability Program for System and Equipment Development and Production

MIL-STD-794 Parts and Equipment, Procedures for Packaging and Packing of

2.1.1 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic devices shall meet the requirements of AR-5.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.2.2 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with Specification AR-10 except as specified herein.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400, for design, construction, and workmanship as otherwise specified herein.

*3.3.1 Total Weight - The total weight of the equipment, excluding cables and interconnecting waveguide, shall not exceed 296 pounds.

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3.3.2 Reliability - The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract.

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of at least 125 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment, including any Built-In-Test Provisions, shall have 125 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - All units of the equipment shall contain time totalizing meters in accordance with Specification MIL-M-7793. The time totalizing meter will be a 10,000 hour meter and will meet military standard MS17321 or MS17322.

3.3.3 Cabling and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.1 Interconnecting Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of individual wires may be shielded when demonstrated as necessary to meet the interference control requirements, and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied with the equipment.

3.3.4 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461, MIL-STD-462, and MIL-STD-463.

3.3.6 Provisions for Maintainability - The maintainability program, built-in-test features, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in Specification AR-10 except as modified in this specification.

3.3.6.1 Compatibility with VAST - Except as specified here-in, the equipment shall be compatible with the Versatile Avionic Shop Test (VAST) system and shall meet the requirements of Specification AR-8.

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3.3.6.2 VAST Test Program - When required by contract, VAST Test Programs shall be furnished in accordance with Specification AR-9. If VAST Test Programs exist for the equipment, and changes to the equipment are made which will affect the fault diagnosis procedure, changes to the existing Test Program shall be prepared as part of the equipment changes in accordance with Specification AR-9.

3.3.6.3 Maintenance Concept - The AN/APS-116 Radar maintenance concept is based on a hybrid mixture of VAST and Peculiar Ground Support Equipment (PGSE) that will insure maximum utilization of VAST, and utilize PGSE only to the extent necessary to over come the limitations of VAST.

BIT (Built-In-Test), IFPM (In Flight Performance Monitoring), and the computer controlled diagnostic program will be used to fault isolate to the WRA at the organizational level. The faulty WRA will be removed and replaced with a good unit from stock.

3.3.6.4 Maintainability Design Features

3.3.6.4.1 Quantitative Requirements - The quantitative requirements for organizational maintenance shall be:

(1) Mean Time to Repair (MTTR) - Shall be no greater than thirty (30) minutes.

(2) Maximum Repair Time (MAX 90%) - Shall be no greater than sixty (60) minutes for 90% of organizational maintenance actions.

3.3.6.6 Built-In-Test (BIT) - A BIT system shall be included in the system design.

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3.3.6.6.1 BIT Confidence Level - The BIT system shall have the capability of detecting 90% of the equipment catastrophic failures. The BIT system will include continuous monitoring for determining system readiness in flight and self-test for preflight checks to determine system readiness. A warning device, of the type that will hold the last test result, will be activated on each WRA.

3.3.6.7 Test Points - Test Points shall be provided on the equipment as follows:

Organization Maintenance, no Test Points required.

Intermediate Maintenance, Test Points shall be provided on the WRA in accordance with paragraph 3.5.1 of AR-10.

Shop Repair Test Points shall be provided per AR-10, paragraph 3.5.4.2 only for SRA's designated as "Repairable" SRA's.

3.3.6.7.1 Functional Test Point Arrangement - The test points described in 3.3.6.7 shall be brought out to and be accessible at a front panel connector. Shop Repair test points need not be brought to the surface of the WRA.

3.3.6.7.1 Maintainability Indices for Organizational Maintenance

3.3.6.8.1 Non-Ambiguity (N-A) Ratio - The ratio of the number of probable malfunctions detected and isolated directly to the WRA with built-in-test features and without ambiguity to the total number of probable malfunctions shall not be less than 0.90.

3.3.6.8.2 Fixed Interface (FI) Ratio - The ratio of the number of WRA's which do not require adjustment or trimming at installation in the aircraft to the total number of WRA's shall not be less than 0.75.

3.3.6.9 Maintainability Indices for Intermediate Maintenance

3.3.6.9.1 Shop Fixed Interface (SFI) Ratio - The ratio of the number of Shop Replaceable Assemblies (SRA's) which do not require adjustment or trimming at installation in the WRA to the total number of SRA's shall not be less than 0.80.

3.3.7 Nomenclature and Nameplates - Nomenclature and serial number assignments, nameplate approval and identification markings shall be in accordance with Specification MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

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Temperature	Room Ambient (25°C±10°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient up to 90% Relative Humidity
Input Power Voltage	115±1.0 VRMS, 3Ø, 400±2 Hz

3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class IX equipment except as modified herein:

3.3.9.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve II of Specification MIL-E-5400.

3.3.9.2 Altitude-Service and Test - Service and Test altitude shall be 40,000 feet maximum.

3.3.9.3 Shock - Shock tests, as specified in MIL-E-5400, are not applicable since vibration isolators are not supplied as part of this equipment.

3.3.10 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed five minutes under standard conditions and 25 minutes at extreme service conditions.

3.3.11 Inputs Electrical Power

3.3.11.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power source having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amount. AC power (three phase), 115/200, Category B, 6000 VA.

3.3.11.2 Degraded Performance - Degraded performance will be permitted for all voltage transients shown on Figure 3 of MIL-STD-704 for Normal Electric System Operation except performance shall not be affected by short pulses of less than 50 microseconds. Recycling the system, cycle from OPERATE TO STBY to OPERATE, will be permissible in order to resume normal operation when the transient expires.

3.3.12 Standby Provisions - The equipment shall have standby provisions whereby rated voltage shall be applied to all tube filaments and circuits requiring a warm-up period. Except for the associated filament circuits and low voltage supplies, high voltages shall not be applied to other parts of the equipment.

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3.3.13 Cooling - Category II and III equipment as defined in MIL-E-5400 shall utilize forced air suction cooling.

3.3.13.1 Category II - Each WRA of a unit in Category II shall have been evaluated for thermal performance in accordance with MIL-T-23103 under a previous contract. This evaluation shall be used in determining the operating temperature check point for Category III units as specified herein.

*3.3.13.2 Cooling Air Pressure Drop - The cooling air pressure drop between the inlet and outlet of the equipment shall be 1 ± 0.2 inches of water at sea level standard day (15°C), at 6 lb/min./KW. The pressure drop in the Transmitter WRA shall be based on the power dissipated when the waveguide switch is in the full radiate position; i.e., no power in dummy load.

Condition	Inlet Air Temperature	Flow Rate
Ambient Steady State, Atmospheric, -54°C	-7°C	14.8 lb/min/kw
Ambient Steady State, Atmospheric, +55°C	+45°C	14.8 lb/min/kw
Ambient Steady State, Altitude, -54°C	-7°C	3 lb/min/kw
Transient, Atmospheric, +71°C	+60°C	23.7 lb/min/kw

COOLING AIR REQUIREMENTS, CATEGORY II

TABLE I

3.3.13.3 Category III - Each Weapons Replaceable Assembly (WRA) shall have a point accessible on the outer surface of the equipment for monitoring the temperature(s) of the most critical part(s) inside. The operating temperatures of the parts selected for the check point shall have been determined from the evaluation of the Category II unit by MIL-T-23103. This check point shall be clearly marked. The installation instruction shall be such that the temperature of the check point shall be maintained within ± 6°C.

3.3.14 Waveguide Pressurization - All waveguide and components requiring pressurization shall be part of a common pressurization system. All components of the pressurized system shall be capable of safely withstanding 30 psig. The maximum rate of pressure loss shall not exceed six pounds in four hours when starting from an initial pressure of 15 psig. This leakage rate shall be based on the standard conditions as specified in paragraph 3.3.8. Freon 116 shall be used as the pressurized gas. The waveguide system shall be pressurized to 26 ± 4 psig. The means to supply the required pressurization shall not be considered part of this equipment.

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3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment shall provide capability for radar detection and processing for station keeping, mine laying, navigation and surface search. It shall provide for periscope and snorkel search in high sea states for target exposure times down to five seconds to ranges of 20 n.miles. Operational modes are:

(1) "Mode I" - High resolution (pulse compression) and high PRF primarily for periscope search.

(2) "Mode II" - Low resolution (0.5 usec transmitted pulse) and low PRF utilizing pulse-to-pulse frequency agility primarily for long range search and navigation.

(3) "Mode III" - High resolution (pulse compression) and low PRF primarily for long range navigation and target classification.

3.4.2 Frequency Characteristics - The Radar operating frequency shall be in the 9.5 to 10.0 GHz band. Transmitter bandwidth shall be consistent with the pulse compression requirements specified in paragraph 3.4.5. Pulse-to-pulse frequency agility in a pseudorandom manner shall be used in "Mode II" with a minimum frequency spread of 300 MHz. The objectives of WR-104 shall be met whenever technically feasible and a report made to the procuring activity of all instances where it is deemed not feasible to meet those objectives. The overall frequency and amplitude characteristics shall be compatible with the pulse compression performance specified in paragraph 3.4.5. The transmitted 3db bandwidth in "Mode I" and "Mode III" shall be 440 MHz minimum.

*3.4.3 Modulation Characteristics - The Transmitter shall be modulated at pulse repetition rates of 500 ± 25 pps for "Mode II" and "Mode III" modes and 1950 ± 70 pps for "Mode I" mode. Transmitted pulsewidth shall be 0.5 usec minimum in all modes.

3.4.4 Transmitter Power Output - The pulse energy at the Transmitter unit output flange shall be no less than 0.22 watt-second for all modes.

*3.4.5 Pulse Compression - The technique used shall consist of a linear FM chirp with receiver weighting for time sidelobe control. The final compressed video 3 dB pulsewidth shall not exceed 3 nanoseconds and the peak-to-peak sidelobe ratio shall be at least 22 dB down under standard conditions. The 3db pulsewidth shall not exceed 3.5 nanoseconds at extreme service conditions.

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3.4.6 Receiver Sensitivity - The maximum Receiver-Pulse Compressor sensitivity shall be equal to or greater than the values specified in the follwoing table:

Mode	P_D	P_{fa}	Receiver-Pulse Compressor Sensitivity
I & III	0.5	10^{-5}	-88.0 dBm
I & III	0.5	10^{-3}	-90.5 dBm
II	Minimum Discernable Signal		-99 dBm

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3.5 Detailed Design Requirements

3.5.1 Power Supply PP-6633/APS-116 - The Power Supply shall meet the following requirements:

3.5.1.1 Functions - The Power Supply shall provide the following functions:

3.5.1.1.1 Primary Power Control - The Radar primary power control circuits shall be contained in the Power Supply unit.

3.5.1.1.2 Regulated DC Supplies - General use unregulated DC voltages for other Radar units shall be provided by the Power Supply unit. Regulation shall be provided in the individual units.

3.5.1.1.3 Servo Drive - The Power Supply shall provide the servo drive and waveforms necessary to scan the S-3A Antenna through all operating modes.

3.5.1.1.4 BIT Reset - The unit shall provide for a pushbutton switch to reset BIT indicators on all Radar WRA's, and three auxilliary outputs for resetting associated equipment BIT indicators.

3.5.1.2 Form Factor - The maximum volume of the Power Supply shall be 1.12 cubic feet. Maximum dimensions are width 11.52 in., height 8.77 in., and depth 19.02 in.

*3.5.1.3 Weight - The maximum weight of the Power Supply shall be 44.0 pounds.

3.5.1.4 Contents - The Power Supply shall contain low voltage regulated power supplies for use within the unit, unregulated DC power supplies for use by the other Radar units, servo amplifiers for each of the four Antenna gimbals, and BIT circuits.

3.5.1.5 Electrical Inputs from other Aircraft Systems

*3.5.1.5.1 A-C Power

115 VAC, 400 Hz, 10 amps per phase.
Limits per MIL-STD-704, Cat. B.

3.5.1.5.2 Pitch Position Input:

Description: 3 wire resolver information referenced to Phase A.

Load: Kearfott CM01037103 resolver or open circuit.

3.5.1.5.3 Roll Position Input:

Description: 3 wire resolver information referenced to Phase A.

Load: Kearfott CM01037103 resolver or open circuit.

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3.5.1.5.4 Synchro Reference

Description: 26 VAC, 400 Hz, referenced to Phase A, derived from MIL-STD-704, Cat. B power.

Load: 0.4 amps maximum.

3.5.1.5.5 Antenna Interlock

Description: Continuity supplied through two connections to enable azimuth servo amplifier when antenna is connected. Lack of continuity inhibits azimuth servo amplifier.

Load: 0.1 amp maximum.

3.5.1.5.6 Pitch Position Error

Description: Kearfott CM01037103 resolver. Rotor output proportional to pitch gimbal position error. Signal referenced to Phase A.

Load: 4,700 OHMS minimum.

3.5.1.5.7 Pitch Rate

Description: Tach generator output proportional to speed of antenna pitch motor (refer to TI Drawing No. 716395).
Scale Factor = 0.3 volts/1000 RPM, referenced to Phase A.

Load: 4,700 OHMS minimum.

3.5.1.5.8 Roll Position Error

Description: Kearfott CM01037103 resolver rotor output proportional to roll gimbal position error. Signal referenced to Phase A.

Load: 4,700 OHMS minimum.

3.5.1.5.9 Roll Rate

Description: Tach generator output proportional to speed of antenna roll motor (Refer to TI Drawing No. 716396).
Scale Factor = 0.3 volts/1000 RPM, referenced to Phase A.

Load: 4,700 OHMS minimum.

3.5.1.5.10 Tilt Position Error

Description: Buweps Type 26V-08CT4C, CT rotor output proportional to tilt gimbal position error. Signal referenced to Phase A.

Load: 4,700 OHMS minimum.

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3.5.1.5.11 Tilt Rate

Description: Tach generator output proportional to speed of antenna tilt motor (refer to TI Drawing No. 716472). Scale Factor = 0.3 volts/1000 RPM.

Load: 4,700 OHMS minimum.

3.5.1.5.12 Azimuth Position Error

Description: Buweps Type 26V-08CT4C, CT rotor output proportional to antenna azimuth position error. Signal referenced to Phase A.

Load: 4,700 OHMS minimum.

*3.5.1.5.13 Azimuth Rate

Description: Tach generator output proportional to speed of antenna gimbal (Refer to TI Drawing 716523).

Scale Factor = 0.65 volts/1000 RPM, referenced to Phase A.

Load: 4,700 OHMS minimum.

*3.5.1.5.14 Antenna Scan Override

Description: Continuity supplied through two input connections when radome is open to inhibit antenna scan. Open circuit supplied when radome is closed.

Load: 0.1 AMP maximum.

3.5.1.5.15 Antenna Stabilization In/Out Command

Description: Stabilization In - Logical "1"
Stabilization Out - Logical "0".

Load: Line Receiver (TI SN 55107).

3.5.1.5.16 Antenna Scan Mode Select

Description: 2 Bit Control word coded as follows:

Antenna	Line A	Line B
STOP	Logical "0"	Logical "0"
SECTOR SCAN	Logical "1"	Logical "0"
SLOW SCAN	Logical "0"	Logical "1"
FAST SCAN	Logical "1"	Logical "1"

Load: Line Receiver (TI SN 55107).

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3.5.1.5.17 Antenna Sector Width Control

Description: Logic Level "1" commands clockwise rotation. Logic Level "0" commands counterwise rotation.

Load: Line Receiver (TI SN 55107).

3.5.1.5.18 Radar ON/OFF Command

Description: -12 VDC denotes radar ON.
-15 VDC through 1 meg. OHM minimum denotes radar OFF.

Load: Relay Coil (300 OHMS minimum).

3.5.1.5.19 Radar Mode Select

Description: 2 bit control word coded as follows:

Radar Mode	Freq	PRF
STBY	Logical "0"	"1"
MODE I	Logical "1"	"1"
MODE II	Logical "0"	"0"
MODE III	Logical "1"	"0"

Load: Line Receiver (TI SN 55107).

3.5.1.5.20 Radar/Antenna Status Transfer Clock

Description: 15 gated square-wave clock pulses. Normally high when inactive. Serial data bits are shifted on each positive excursion of clock pulses.

Load: Line Receiver (TI SN 55107).

3.5.1.5.21 Radar/Antenna Status Transfer Gate

Description: Logic Gate. Positive edge starts transfer on next clock pulse.

Load: Line Receiver (TI SN 55107).

3.5.1.5.22 Encoder Fail Indication

Description: Logical "1" denotes encoder fail.

Load: Line Receiver (TI SN 55107).

3.5.1.6 Electrical Outputs to other Aircraft Systems

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3.5.1.6.1 Radar Warm-Up Indication

Description: Logical "1" denotes radar warm-up is complete.

Load: Line Receiver (TI SN 55107).

3.5.1.6.2 Fault Indicator Resets

Description: Momentary (0.5 seconds) or continuous 28 VDC to reset fault indicators. Three individual outputs shall be provided.

Load: Relay Coils (1000 OHMS minimum for each output).

3.5.1.6.3 WRA Status Word

Description: 15-bit serial word. Logical "0" denotes fail. Spare bits snall be logical "1". Serial data bits shift on each positive excursion of the input clock pulses.

Bit	Status Signal
1	Antenna Roll Gimbal
2	Antenna Pitch Gimbal
3	Antenna Tilt Gimbal
4	Antenna Azimuth Gimbal
6	Synchronizer-Exciter
7	Transmitter
8	Receiver-Pulse Compressor
9	Antenna Encoder
10	Antenna
11-15	Spares

Load: Line Receiver (TI SN 55107).

3.5.1.6.4 26 VAC Synchro Reference

Description: 26 VAC, 400 Hz referenced to Phase A. Derived from MIL-STD-704, Cat. B power. Available when power supply is connected.

Load: 0.1 amps maximum.

3.5.1.6.5 Roll Stabilization

Description: 3 wire resolver output referenced to Phase A.

Load: Kearfott CM01037103.

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3.5.1.6.6 Pitch Stabilization

Description: 3 wire resolver output referenced to Phase A.

Load: Kearfott CM01037103.

*3.5.1.6.7 Azimuth Drive

Description: A-C Drive Voltage. Refer to TI Drawing 716523 for detailed specifications.

3.5.1.6.8 Pitch Drive

Description: A-C Drive Voltage. Refer to TI Drawing 716395 for detailed specifications.

3.5.1.6.9 Roll Drive

Description: A-C Drive Voltage. Refer to TI Drawing 716396 for detailed specification.

3.5.1.6.10 Tilt Drive

Description: A-C Drive Voltage. Refer to TI Drawing 716472 for detailed specifications.

3.5.1.6.11 Stow Brakes Release

Description: 28 VDC supplied for Stow Brake release.

Load: 0.25 amps maximum.

*3.5.1.6.12 Antenna Primary Power

Description: 115 VAC, 400 Hz, 3 phase, 4 wire supplied from MIL-STD-704, Cat. B, power.

Load: 0.5 amps per phase maximum.

3.5.1.6.13 Antenna Fault Indicator Drive

Description: 28 VDC supplied to denote antenna failure.

Load: Relay Coil (1000 OHMS minimum).

3.5.1.6.14 26 VAC Synchro Reference

Description: 26 VAC, 400 Hz, referenced to Phase A. Derived from MIL-STD-704, Cat. B power. Available when radar is energized and connected to the Antenna.

Load: 0.1 amps maximum.

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3.5.2 Synchronizer-Exciter SN-460/APS-116 - The Synchronizer-Exciter shall meet the following requirements:

3.5.2.1 Function - The Synchronizer-Exciter shall provide the following functions:

3.5.2.1.1 Triggers - The Synchronizer-Exciter shall provide all necessary synchronization triggers for the Radar. A reference clock shall be provided for use by other units which is phase locked with respect to output triggers.

3.5.2.1.2 Local Oscillator - The Synchronizer-Exciter shall provide the local oscillator signals required by the Receiver-Pulse Compressor in all modes of operation.

3.5.2.1.3 Transmitter Signal - The Synchronizer-Exciter shall provide the pulsed RF signal input to the Transmitter. Appropriate FM and AM modulation shall be included upon receipt by the Synchronizer-Exciter of mode control signals.

3.5.2.1.4 Test Signal - A test signal shall be generated prior to transmit time that will allow self-test of critical equipment functions.

3.5.2.2 Form Factor - The maximum volume of the Synchronizer-Exciter shall be 0.96 cubic feet. Maximum dimensions are width 10.02 in., height 9.46 in., and depth 17.02 in.

*3.5.2.3 Weight - The maximum weight of the Synchronizer-Exciter shall be 36.0 pounds.

3.5.2.4 Contents - The Synchronization-Exciter shall contain the reference clock, synchronizer, master oscillator and transmitter lock loops, TWT amplifier and BIT circuits.

3.5.2.5 Electrical Inputs from other Aircraft Systems

3.5.2.5.1 IFF Interrogator ON/OFF

Description: 28 VDC Denotes IFF Interrogate ON.

Load: Relay Coil (500 OHMS minimum).

3.5.2.5.2 RF Radiation Inhibit

Description: Logical "1" corresponds to inhibit RF drive condition.

Load: Line Driver (TI SN 55107).

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3.5.2.5.3 Trigger from IFF Interrogator

Description:		
	Polarity	Positive
	Amplitude	+50 ± 10 volts
	Width	2.2 ± 1 μsec.
	Rise Time	0.1 μ sec. max.
	Fall Time	1.5 μ sec. max.
	DC Level	0 ± 1 volt
	Delay	462 ± 0.5 μ sec. after trigger to IFF interrogator

Load: 91 OHMS.

3.5.2.6 Electrical Outputs to other Aircraft Systems

*3.5.2.6.1 Trigger to IFF Interrogator

Description:		
	Polarity	Positive
	Amplitude	+15 to +60 volts
	Rise Time	0.5 μ sec. max.
	Fall Time	1.0 μ sec. max.
	Width	0.5 to 5.0 μ sec.
	Repetition Rate	500 ± 25 or 1950 ± 70 pps
	DC Level	0 ± 1.0 volts

Load: 91 OHMS.

*3.5.2.6.2 Blanking Pulse

Description:		
	Polarity	Positive
	Amplitude	4.5 ± 0.5 volts
	Rise Time	0.1 μ sec. max.
	Fall Time	0.1 μ sec. max.
	Width	2.75 ± 0.20 μsec.
	Timing	Pulse starts 1.1 ± 0.1 μsec. in advance of transmit pulse.

Load: 91 OHMS.

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3.5.2.6.3 Zero Range Trigger

Description: Pulsewidth 240 sec ± 40 μ sec
Delay Time: 840 sec ± 40 μ sec from Master Trigger (leading edge to leading edge)

Load: Line Receiver (TI SN 55107).

3.5.2.6.4 RIU Pretrigger

Description: Pulsewidth: 240 n sec ± 40 n sec

Timing: 30.84 μsec in advance of master trigger (leading edge to leading edge)

Load: Line Receiver (TI SN 55107).

3.5.2.6.5 8.33 MHz Clock

Description: 8.33 MHz square wave.

Load: Line Receiver (TI SN 55107)

3.5.3 Transmitter, Radar Set T-1203/APS-116 - The Transmitter shall meet the following requirements:

3.5.3.1 Function - The Transmitter shall consist of a pulsed amplifier chain and appropriate modulators. It shall accept RF drive from the Synchronizer-Exciter and amplify it to a level conforming with the requirements of paragraphs 3.4.4.

*3.5.3.1.1 Output Power Control - The transmitter shall include a waveguide switch to allow transmitter power to be directed to either the transmitter output port or into the transmitter dummy load in one step using external control signals.

3.5.3.1.2 Intermittent Operation - The transmitter shall allow intermittent operation to minimize ECM alerting.

*3.5.3.2 Form Factor - The maximum volume of the Transmitter shall be 3.30 cubic feet. Maximum dimensions are: width 21.82 in., height 12.59 in., and depth 20.69 in.

*3.5.3.3 Weight - The maximum weight of the Transmitter shall be 174 pounds.

3.5.3.4 Contents - The Transmitter shall contain a TWT amplifier and modulator, a CFA amplifier and modulator, a variable power driver, a load isolator, a heat exchanger and BIT circuits.

3.5.3.5 Electrical Inputs from other Aircraft Systems

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3.5.3.5.1 RF Radiation Inhibit

Description: Logical "1" corresponds to inhibit RF drive condition.

Load: Line Receiver (TI SN 55107).

*3.5.3.5.2 Radiated Power Control

Description: DC Control Voltage Balanced about ground.

Differential Voltage Range	Power Direction
0 ± 0.5 volts DC	Transmitter Output Port
5 $^{+1}_{-0}$ Volts DC	Transmitter Dummy Load

Load: 5000 ohms, nominal.

3.5.3.5.3 Transmitter Primary Power

Description: 115 VAC, 400 Hz, 3 Phase, 4 wire, Cat. B., limits per MIL-STD-704.

Load: 11 amps maximum.

*3.5.3.5.4 RF Output Override

Description: Continuity supplied between two input pins when aircraft is on ground.

Load: 0.1 AMP maximum.

3.5.3.6 Electrical Outputs to other Aircraft Systems

3.5.3.6.1 HV ON Indication

Description: Logical "1" denotes transmitter HV ON.

Load: Line Receiver (TI SN 55107).

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3.5.3.6.2 RF ON Indication

Description: Logical "1" denotes transmitter RF ON.

Load: Line Receiver (TI SN 55107).

3.5.3.7 Distortion - The phase modulation, amplitude modulation, instantaneous bandwidth, dynamic range, and modulator timing jitter shall be consistent with the pulse compression requirements as specified in paragraph 3.4.5.

3.5.3.8 Pressurization - The Transmitter waveguide elements shall be pressurized. The leakage rate shall be consistent with paragraph 3.3.14.

3.5.4 Receiver-Pulse Compressor R-1747/APS-116 - The Receiver-Pulse Compressor shall meet the following requirements:

3.5.4.1 Function - The Receiver-Pulse Compressor shall provide the following functions:

3.5.4.1.1 Receiver-Pulse Compressor Functions - The Receiver-Pulse Compressor shall possess an instantaneous bandwidth and a phase modulation characteristic consistent with the pulse compression requirements as specified in 3.4.5. It shall incorporate suitable selectivity prior to any amplifying stage to minimize mutual interference and off-frequency jamming. The compression network shall provide a linear delay vs. frequency response which is the conjugate of the waveform generated in the Synchronizer-Exciter. Suitable weighting shall be provided to meet the requirements of 3.4.5. Sufficient IF gain shall be provided to achieve the required signal level prior to video detection. Video processing shall provide an adjustable false alarm rate which is relatively independent of range and azimuth in both the clutter and noise limited region. The prime objective shall be to maximize overall detection performance at all points of the search area.

3.5.4.1.2 AGC Function - The Receiver-Pulse Compressor shall provide means for pre-compression gain control to limit the dynamic range of signals prior to pulse compression.

3.5.4.2 Form Factor - The maximum volume of the Receiver-Pulse Compressor shall be 1.30 cubic feet. The maximum dimensions are width 13.02 in., height 10.52 in., and depth 16.02 in.

*3.5.4.3 Weight - The maximum weight of the Receiver-Pulse Compressor shall be 48.0 pounds.

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3.5.4.4 Contents - The Receiver-Pulse Compressor shall contain the duplexer, RF amplifier, down converter, IF amplifiers, AGC circuits, pulse compression filter, weighting filters, detector, and threshold circuits for "Mode I" mode of operation.

3.5.4.5 Electrical Inputs from other Aircraft Systems

3.5.4.5.1 Received RF Energy

Description: Received RF energy per this specification requirements.

Load: Duplexer.

3.5.4.5.2 Radar Display Range

Description: Logical "1" denotes a display range greater than 32 N. miles. Logical "0" denotes display range less than 32 N. miles.

Load: Line Receiver (TI SN 55107).

3.5.4.5.3 Receiver Gain Control

Description: DC Control Voltage balanced about ground

Level	Gain	Differential Voltage
8	Maximum	0 ± 0.1 volts
7		↑
6		|
5		Linear Change
4		|
3		⋮
2		↓
1	Minimum	7 ± 0.1 volts

Load: 1000 OHMS minimum.

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3.5.4.5.4 PPI FAR Control

Description: DC Control Voltage balanced about ground

Level	FAR	Differential Voltage
1	Minimum	0 ± 0.1 volts
2		↑
3		↑
4		↑
5		Linear Change
6		↓
7		↓
8	Maximum	7 ± 0.1 volts

Load: 1000 OHMS minimum.

3.5.4.5.5 B-Scan FAR Control

Description: DC Control Voltage balanced about ground

Level	FAR	Differential Voltage
1	Minimum	0 ± 0.1 volts
2		↑
3		↑
4		↑
5		Linear Change
6		↓
7		↓
8	Maximum	7 ± 0.1 volts

Load: 1000 OHMS Minimum.

3.5.4.5.6 RF Radiation Inhibit

Description: Logical "1" corresponds to inhibit RF drive condition.

Load: Line Receiver (TI SN 55107).

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3.5.4.6 Electrical Outputs to other Aircraft Systems

*3.5.4.6.1 Transmit Pulse

Description: Transmitted RF energy per this specification.

Load: Antenna
Maximum VSWR = 1.7:1.

3.5.4.6.2 PPI Threshold Video

Description:	Amplitude	3 ± 0.5 volts
	Pulsewidth	30 ± 10 nsec

Load: 50 OHMS.

3.5.4.6.3 B-Scan Threshold Video

Description:	Amplitude	3 ± 0.5 volts
	Pulsewidth	30 ± 10 nsec

Load: 50 OHMS.

3.5.4.6.4 B-Scan Raw Video

Description:	Amplitude	3 ± 0.5 volts
	Pulsewidth (3 db)	12 to 18 nsec

Load: 50 OHMS.

*3.5.4.6.5 Nav Video

Description:	Amplitude	3 ± 0.5 volts
	Pulsewidth	0.4 to 0.6 usec.
	RMS Noise	1.0 volt minimum (Rec. gain max.)

Load: 50 OHMS.

3.5.4.7 Pressurization - The duplexer shall be pressurized in accordance with 3.3.14.

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4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified, the supplier is responsible for the performance of all test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications, and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

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4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection of equipment already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on the equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause of deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the Initial Production Test the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity.

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The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacturing shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Test

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least ten hours of operation.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ± 3g's.

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The equipment shall be vibrated (without vibration isolators) for a period of ten minutes prior to the beginning of the ten hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the ten hour period of operation following the ten minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The ten hour period specified above may be composed of two five hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following:

QUANTITY OF EQUIPMENTS OFFERED FOR ACCEPTANCE	QUANTITY TO BE SELECTED FOR SAMPLING TEST
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof.

One out of first ten need not be selected and tested if initial production tests are conducted. Sampling tests are not required when reliability assurance tests are conducted.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational tests at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

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(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the Preproduction, Initial Production, Individual and Special Tests.

(3) Manufacturing Run In Test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce possibility of its recurrance in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level "E" shall be used. The Accept-Reject Criteria for Test Plan IV shall be used.

4.4.3.2 Reliability Production Acceptance (Sampling) Phase Tests - The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests". Test level "E", Test Plan XXIX, shall be used.

4.4.3.2.1 All Equipment Test - Each equipment produced except those submitted for the Reliability Qualification Test, shall be tested for 50 hours. Prior to the 50 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used, the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data). These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a

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reject situation exists, pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject situation.

4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change

(2) Whenever failure reports or other information indicate that additional tests are required (this will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of the failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failures occurring in future tests.

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(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.). When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
first 25	1
next 175	1
next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000ft.)
Humidity	Room ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately eight hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

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4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production, acceptance and life tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

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4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and re-submitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with the Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - This specification defines the Radar Set, AN/APS-116( ) for the procurement of production models as defined in MIL-E-5400.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification

(2) Selection of applicable levels of packaging and packing (See 5.1)

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

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(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.7. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the asterisk notations and relationship to the last previous issue.

6.7 Associated Equipment - The AN/APS-116 shall operate with the following accessories and associated equipment:

(1) IFF Interrogator Set, AN/APX-76, MIL-I-84153(AS)
(2) Mountings - Mounting bases shall not be included as part of this equipment.
(3) Aircraft Cooling System - See Paragraph 3.3.13 for requirements.
(4) S-3A Radar Antenna
(5) Converter-Storer, Signal Data CV-2852/AP
(6) Control, Radar Set, C-8788/AP

*6.8 Cognizance - This specification is under the cognizance of NAVAIR Code 533318.

Preparing Activity - NAVY AS

Project Number 5841- NO64

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DOCUMENT IDENTIFIER AND TITLE
MIL-R-81648B(AS) Radar Set AN/APS=116

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